

Wilson Costa

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Professional Experience

BMW - Critical Techworks - Devoteam

February 2023 – Present

Backend Engineer and Scrum Master

- Served as Backend Engineer and Scrum Master on a **large-scale distributed system** using Quarkus services, each with dedicated PostgreSQL schemas. **Infrastructure management** with Kubernetes deployments using Helmfile, CI/CD pipelines with GitHub Actions, infrastructure creation with Terragrunt + Terraform for Azure. **Operations** monitoring with Grafana dashboards, and support to ITSM ticket handling using ServiceNow.
- **Led Kafka-based data migration**, creating a microservice to process thousands of large JSON events (~2MB each); implemented parallelization and topic partitioning, turning **10K events, 15s/event** batch process into an asynchronous near-real-time pipeline.
- **Optimized monolith imports**, improving in-memory graph usage and fixing Hibernate queries, having more than **100% improvement** by reducing processing time from **~20 minutes** to **~15 seconds**. Achieved by establishing the correct associations between entities and enhancing the search mechanisms using hash based traversal.
- **Boosted Database Performance**: Increased PostgreSQL performance by indexing and rewriting queries, reducing operations from **~1/2 minutes** to **500ms** across partitioned tables with **10+ millions** of rows.
- **Quarkus Migration Update**: Executed migrations from **Quarkus 2 → 3.20** using structured upgrade recipe templates with Maven Rewrites to ensure reliability and compatibility, while creating an **automated migration** process.
- **Enhanced Frontend Suggestions**: Improved Monaco Editor having **10K+** auto-suggestions in Angular from **30s per keystroke** to **~10ms**, enabling real-time user experience.
- **Built distributed caching**: Developed a **headless distributed cache** using Hazelcast in Quarkus, replacing service-specific Caffeine caches with a centralized cache managed by a primary service, ensuring cache updates are **propagated immediately** across all services instead of waiting for individual cache refreshes.
- **Augmented Security Integration**: Implemented **Quarkus OIDC-based authentication**, enriching user personal information coming from the OIDC provider with external-provider roles.
- **Agile Leadership**: Facilitated standups, planning, retros, and JIRA scope management as **Scrum Master**, while mentoring other developers.

BMW - Critical Techworks - Devoteam

October 2021 - February 2023

Full Stack Developer

- Delivered fullstack features for a modular widget-based architecture with a dashboard platform with **real-time data** and third-party integrations.
- Built Quarkus and NestJS APIs, exposing data via GraphQL, CosmosDB (cloud NoSQL) and PostgreSQL for data persistence.
- Managed Kubernetes deployments via Helmfile, Jenkins and ArgoCD pipelines for automated builds and releases.
- **Built Widget Ecosystem Development**: Built modular, reusable widgets (e.g. sales timelines, world clock, car descriptions, visualization of car errors, and Microsoft Teams integration) supported by APIs, improving dashboard extensibility.
- **Integration of a Content Management System**: Implemented a news side panel integrated with Strapi as the CMS, enabling designated users to create and manage news articles, which are dynamically displayed in the side panel.
- **Designed High-Performance APIs**: The backend microservices achieved **<1 second response times**, enabling real-time widget updates.
- **Database Engineering**: Developed and optimized queries in **CosmosDB and PostgreSQL**, improving data reliability and supporting high-traffic dashboard usage for **1000+** users.
- **Frontend Migration Update**: Applied an Angular migration from **version 12** to **14** and a BMW internal library using the latest components and styles.

Morphis Tech - Devoteam

Full Stack Developer

March 2021 - September 2021

- Worked on modernizing legacy **Oracle Forms applications** into web-based systems, improving maintainability, performance, and frontend-backend integration.
- Used AngularJS for a **legacy modernization** of user experience and an interactive web UI. Groovy on Grails for server-side logic. Oracle SQL for data persistence. GitLab and Jenkins for source control and automated deployments.
- **Data Modernization & Integration:** Recreated the **data creation workflow using JSON**, enabling seamless communication with AngularJS front-end.
- **Business Logic Optimization:** Refactored legacy **JavaScript tax functions** to be simpler, more efficient, and reliable.
- **Performance Improvements:** Reviewed auto-transformed APIs and **optimized database access**, significantly reducing query times and improving system responsiveness.

Education

Instituto Superior de Engenharia, Lisbon (ISEL)

Master's Degree in Informatics and Computer Engineering (only 1st year)

September 2021

Focus: Advanced Software Engineering, Distributed Systems, Cybersecurity

Main Projects:

- *Travel Planner App* + *API* (Software Engineering): Angular, Spring Boot - Java, PostgreSQL, Kubernetes Deployments, Google Authentication, Trip Advisor API
- *CES Vector* (Distributed Systems): Transaction management system (Quarkus, Kubernetes, Docker)

Bachelor's Degree in Informatics and Computer Engineering

October 2020

Focus: Programming, Web Development, Information Systems, Cloud Computing, Security, Networking

Main Projects:

- *Invoice Manager* (BSc Final Project): Laravel API + React frontend with OCR integration using Tesseract
- *Text System* (Cloud Computing): Java-based system on GCP using Vision & Translation APIs with dynamic VM allocation
- *Project Manager* (Web Development): React + Sprint Boot - Kotlin + PostgreSQL
- *Ciborg Games* (Internet Programming): Javascript + ExpressJS (Node with Auth)

Skills

Programming Languages: Java, TypeScript, JavaScript, SQL, Kotlin, NodeJS

Frameworks & Tools: Quarkus, Spring Boot, Angular, React, Kafka, Helmfile, Terraform, Terragrunt, Kubernetes, Jenkins, GitHub Actions

Databases: PostgreSQL, CosmosDB, Oracle SQL

Spoken Languages: Portuguese (native), English (fluent), Spanish (basic)

Certifications:

- Professional Scrum Master – Scrum.org | 2024
- AWS Certified Cloud Practitioner – Amazon Web Services | 2022
- Associate Cloud Engineer – Google Cloud | 2021

Personal Projects:

- *Dish Master*: Spring WebFlux and Elasticsearch to create meals.
- *Shining Code*: MERN Stack Application for a clothing e-Commerce Shop.
- *Nutrisport*: Spring Boot API to present food and recipes with nutrition tables.